

◆ DAY 03 OF 30

OneLake Settings & Access Control Architecture

Master Workspace Roles, Item Permissions, OneLake Data Access Roles & Shortcut Security for the DP-700 Exam

The 3 Layers of Fabric Security

Security in Microsoft Fabric operates on a Defense-in-Depth framework spanning workspace management down to file paths.

1 Workspace Roles

Coarse-grained governance controlling who can build, manage, or delete assets inside a workspace.

Roles: Admin, Member, Contributor, Viewer.

2 Item Permissions

Granular item-level access without granting workspace-wide rights.

Permissions: Read, ReadData, Write, Reshare, Execute.

3 OneLake Access Roles

Fine-grained storage path security enforced directly on OneLake folders, files, and Delta tables across all engines.

Layer 1: Workspace Roles Matrix

Understanding privileges across Admin, Member, Contributor, and Viewer workspace roles.

Workspace Privilege	Admin	Member	Contributor	Viewer
Delete Workspace or Modify Capacity	✔ Yes	✘ No	✘ No	✘ No
Add Users / Manage Workspace Roles	✔ Yes	✔ Yes	✘ No	✘ No
Create, Edit & Delete Items (Notebooks/Pipelines)	✔ Yes	✔ Yes	✔ Yes	✘ No
Read Underlying Lakehouse Data (Spark/T-SQL)	✔ Yes	✔ Yes	✔ Yes	⚠ Needs ReadData
View Reports & Visualizations Only	✔ Yes	✔ Yes	✔ Yes	✔ Yes

Layer 2: Item Permissions Deep Dive

Grant access to specific Lakehouses or Data Warehouses without exposing the rest of the workspace.

Read vs. ReadData

Read: Allows opening reports & viewing item metadata.

ReadData: Required to query underlying Delta files via SQL Endpoint or PySpark.

💡 *Formula: SQL Endpoint Access = Read + ReadData.*

Write, Reshare & Execute

Write: Update Delta tables, run DDL/DML, and modify schemas.

Reshare: Share the specific item with external colleagues.

Execute: Trigger pipeline runs and notebook execution.

DAY 03 OneLake Security

Layer 3: OneLake Data Access Roles

Fine-grained, engine-agnostic security directly on OneLake storage paths and Delta tables.

Folder & File Security

Restrict user access to specific subfolders inside the Lakehouse `Files/` directory.

Ideal for raw landing zone data segregation.

Table-Level Security

Grant or restrict access to specific Delta tables inside `Tables/`.

Enforced uniformly across **Spark, SQL Endpoint, and Power BI DirectLake**.

DAY 03 Shortcuts Security

OneLake Shortcut Security Inheritance

How authentication and permissions behave when linking external ADLS Gen2 or AWS S3 data.

User Identity Passthrough

Queries evaluate the individual user's credentials directly against ADLS/S3 IAM policies.

Requirement: End-users MUST have direct storage permissions on the target cloud.

Delegated Organizational Credentials

Queries use a stored Service Principal or organizational connection.

Requirement: Users only need Fabric permissions; no target cloud account needed.

SQL Endpoint vs Spark Security Enforcement

Understanding where RLS, CLS, and Dynamic Data Masking apply.

Security Mechanism	SQL Analytics Endpoint	PySpark Notebooks	Power BI DirectLake
Row-Level Security (RLS)	✔ Enforced via T-SQL	✘ Bypassable (Direct File Read)	✔ Honored via SQL Endpoint
Column-Level Security (CLS)	✔ Enforced via T-SQL	✘ Bypassable (Direct File Read)	✔ Honored via SQL Endpoint
OneLake Data Access Roles	✔ Enforced	✔ Enforced	✔ Enforced

DAY 03 DP-700 Traps

Top 3 DP-700 Security Exam Traps

Avoid these high-frequency exam mistakes on test day.

⚠️ Trap 1: Contributors

Contributors CANNOT add users or manage workspace roles. Only Admins and Members can.

⚠️ Trap 2: Viewers

Workspace Viewers CANNOT run SQL queries unless granted ReadData item permission.

⚠️ Trap 3: SQL RLS

T-SQL RLS does NOT protect files if users have direct Spark read access. Use OneLake Access Roles instead!

Security Decision Cheat Sheet

Quick reference guide for selecting security models in scenario questions.

Business Requirement	Recommended Security Model
Need full control to delete items and manage capacity	Workspace Role: Admin
Need to build pipelines but NOT manage user roles	Workspace Role: Contributor
Allow business user to query SQL Endpoint only	Item Permission: Read + ReadData
Secure Delta tables across Spark, SQL & DirectLake	OneLake Data Access Roles
Expose AWS S3 data to users without AWS IAM accounts	Shortcut with Delegated Credentials

◆ DAY 03 COMPLETE

Follow for Day 04 tomorrow!

Topic: Apache Airflow & Pipeline Orchestration in Microsoft Fabric

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